MIL-R-52002A(ME)
1 December 1971
SUPERSEDING
MIL-R-52002(CE)
5 November 1957

MILITARY SPECIFICATION

RACK ASSEMBLY, CONE, AIRPORT MARKER LIGHT

1. SCOPE

1.1 This specification covers a cone rack assembly for airport marker light cones.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-200/8	-	Aluminum Alloy Bar, Rod, Shapes, Tube and Wire, Extruded, 6061.
QQ-A-225/8	-	Aluminum Alloy Bar, Rod, Wire, and Special Shapes; Rolled, Drawn, or Cold Finished, 6061.
QQ-A-250/11	-	Aluminum Alloy 6061, Plate and Sheet.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Box, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.

Military

MIL-W-530	-	Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-E-15597	-	Electrodes, Welding, Covered, Coated, Aluminum and Aluminum Alloy.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N.Y. 10001.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, DC 20036.)

AMERICAN WELDING SOCIETY, INC. (AWS)

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N.Y. 10017.)

3. REQUIREMENTS

3.1 Description. The cone rack assembly shall consist of a base plate, 2-inch web straps with handling loops, a buckle, and a cap as shown in figures 1, 2, and 3 and as specified herein. The web straps shall have an effective length of 54 inches and shall hold 36 cones firmly when the strap is tightened through the buckle. Cones shall not be furnished under this specification.

3.2 Material. Material shall be as specified herein. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

3.3 Aluminum alloy. Aluminum alloy shall conform to the following specifications:

Form	Specifications
Bars, rods, and shapes	QQ-A-200/8, temper T6, or QQ-A-225/8, temper T6
Sheet and plate	QQ-A-250/11, temper T6

3.3.1 Webbing. The cotton webbing for the strap shall conform to MIL-W-550, type IIa, class 4, olive-drab color.

3.3.2 Welding electrodes. Electric-welding electrodes shall be in accordance with MIL-E-15597. Gas-welding electrodes shall conform to the alloy specified in 3.3.

3.4 Identification marking. Each rack assembly shall be identified in accordance with MIL-STD-130.

3.5 Treatment and painting. The rack assembly shall be treated and painted in accordance with MIL-T-704, type A.

3.6 Workmanship.

3.6.1 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.6.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall develop strength and transmit stress under proof or service loading without failure or permanent deformation. Welding shall be in accordance with a welding procedure for 1/16-inch and 1/8-inch aluminum plate as defined in the American Welding Society handbook.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contractor purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance requirements (see 4.3).
(b) Inspection of preparation for delivery (see 4.5)

4.3 Quality conformance inspection.

4.3.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105. Samples selected shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent for minor defects.

4.4 Inspection procedure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

4.4.1 Examination. The rack shall be examined for the following defects:

Major

101. Nonconformance to figures 1, 2, and 3.
102. Failure of buckle to tighten and release strap.

Minor

201. Defective web straps.
202. Web straps incorrectly fastened or secured to the base, cap, or buckle.
203. Treatment and painting not as specified.
204. Workmanship not as specified.
205. Sharp, ragged, or rough edges.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

103. Materials or containers not as specified for levels A and B.
104. Contents not cushioned as specified for level A or B.
105. Box not waterproof sealed as specified for level A.
106. Strapping not as specified for level A.
107. Gross weight or size exceeds the limitations of the box specification for level B.
108. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified.

5.1.1 Level A. Each complete rack assembly shall be packaged in a close-fitting box conforming to PPP-B-636, W6c, style optional. The contents shall be cushioned to prevent damage and free movement within the box. The box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

5.1.2 Level B. Each complete rack assembly shall be packed as specified in 5.1.1 for level A except the box shall be type CF, class domestic, variethy SW, grade 125, and waterproof sealing of the box shall not be required.

5.1.3 Level C. Each complete rack assembly shall be packaged in a close-fitting fiberboard box adequate to afford protection against protection against damage from the supplier to the initial designation.

5.2 Packing. Packing shall be level A, B, or C as specified.

5.2.1 Level A. The rack assemblies packaged as specified in 5.1 shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

style optional. Box closure and strapping shall be accordance with the appendix to the box specification. Strapping shall be class B.

5.2.2 Level B. The rack assemblies packaged as specified in 5.1 shall be packed together in close-fitting boxes conforming to PPP-B-636, V3c, style optional or PPP-B-640, class 2, style optional. The gross weight or size of each box shall not exceed the limitations of the applicable box specification. Box closure and strapping shall be in accordance with the applicable box specification.

5.2.3 Level C. The rack assemblies packaged as specified in 5.1 shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in a fiberboard container complying with Uniform Freight Classification rules or with National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required in the contract or order, interior package and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The rack assembly is used to hold and transport marker light cones.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2). Level B packaging and packing is intended to provide economical but limited protection and should be specified only when it is determined the rack assemblies will be held in covered storage.

Custodian:
Army - ME

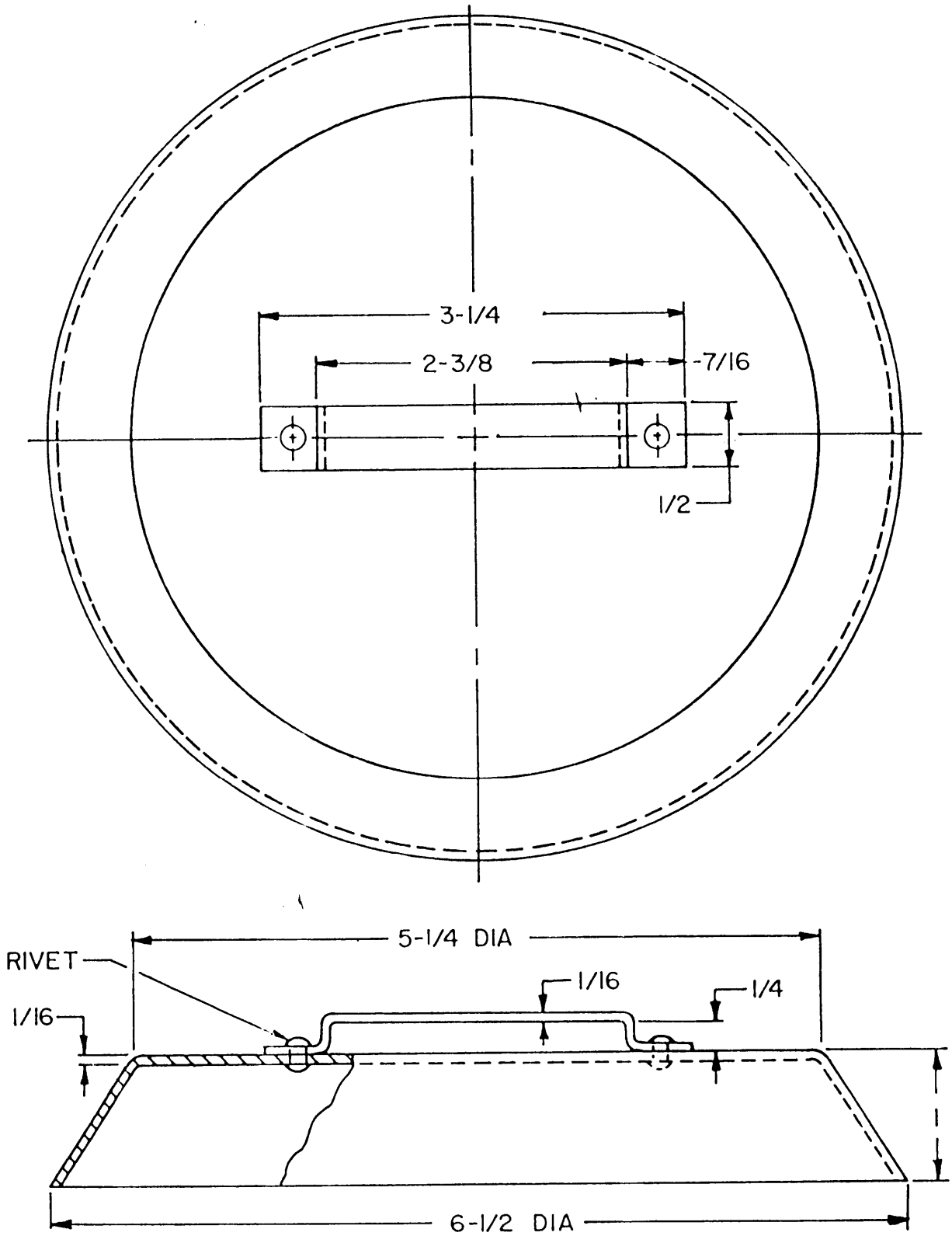
Preparing activity:
Army - ME

Project 6210-A296

User interest:

Army - AV

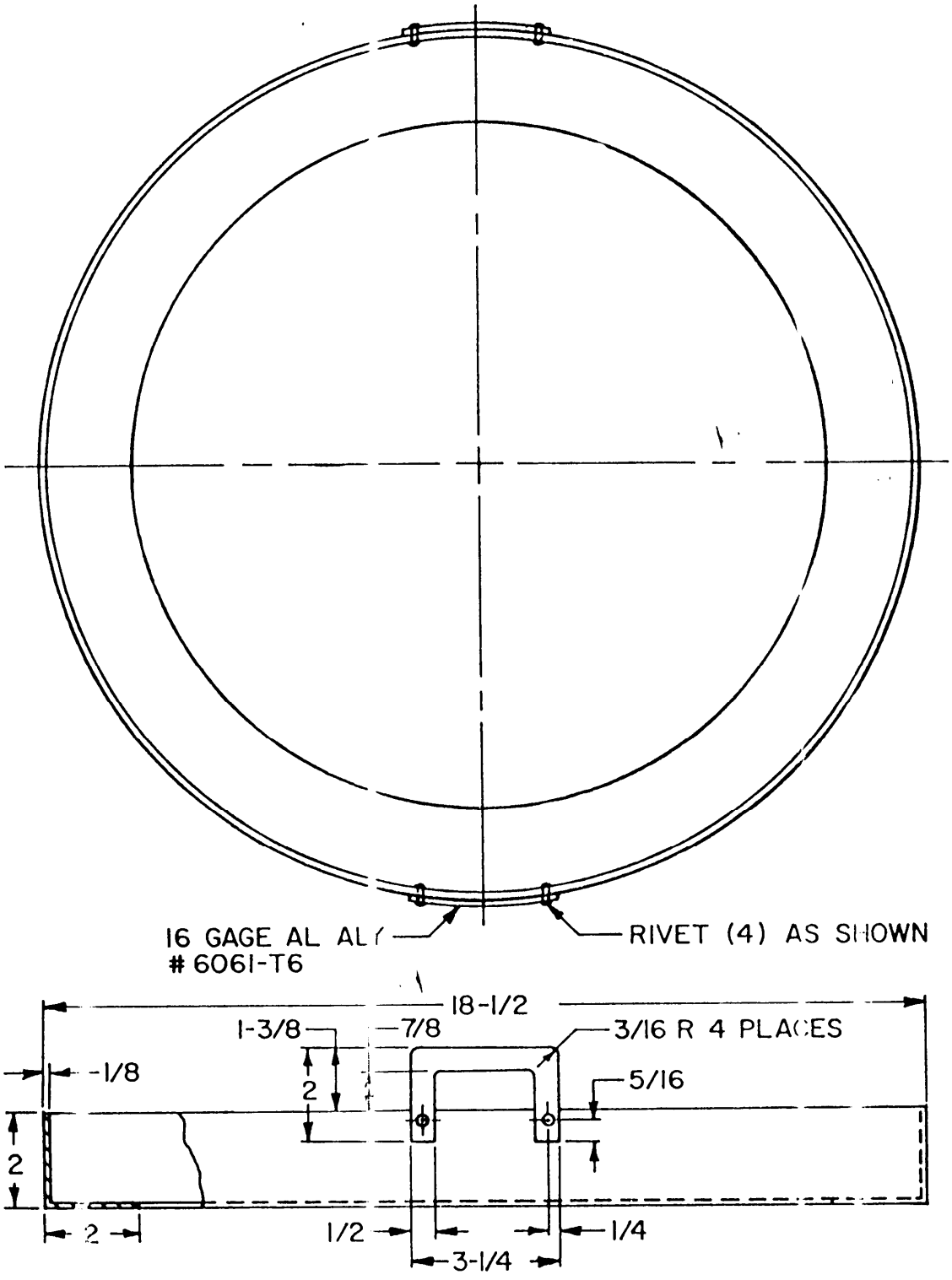
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.



NOTE:
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES AND TOLERANCE IS ± 1/64.

FIGURE 1. CAP ALUMINUM ALLOY A3431- 1A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.



NOTE :
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES AND TOLERANCE IS ± 1/64

FIGURE 2. BASE ALUMINUM ALLOY A3431-2A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

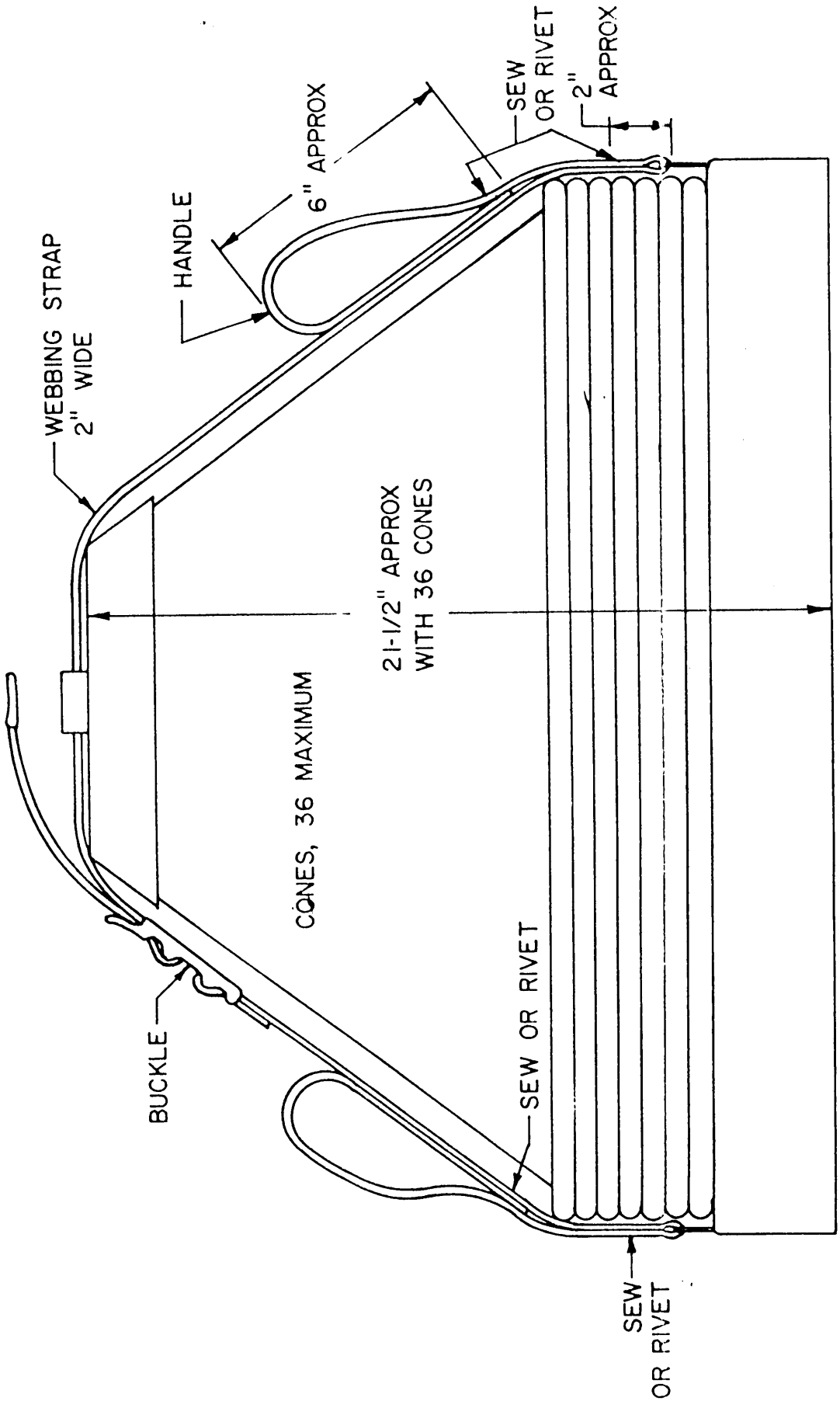


FIGURE 3. CONE RACK ASSEMBLY

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-R-52002A (ME)	**2. DOCUMENT DATE (YYMMDD)** 711201

3. DOCUMENT TITLE
RACK ASSEMBLY, CONE, AIRPORT MARKER LIGHT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME BELVOIR CECOM SOUTH	b. TELEPHONE *Include Area Code)* (1) Commercial (703) 704-3468 (2) AUTOVON 654-3468
c. ADDRESS *(Include Zip Code)* 10115 GRIDLEY ROAD SUITE 228 ATTN: AMSEL-LC-IEW-D-ED FORT BELVOIR, VA 22060-5849	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.